

DETECTION OF AI-GENERATED ARABIC TEXT: A DATA MINING APPROACH

Presented by
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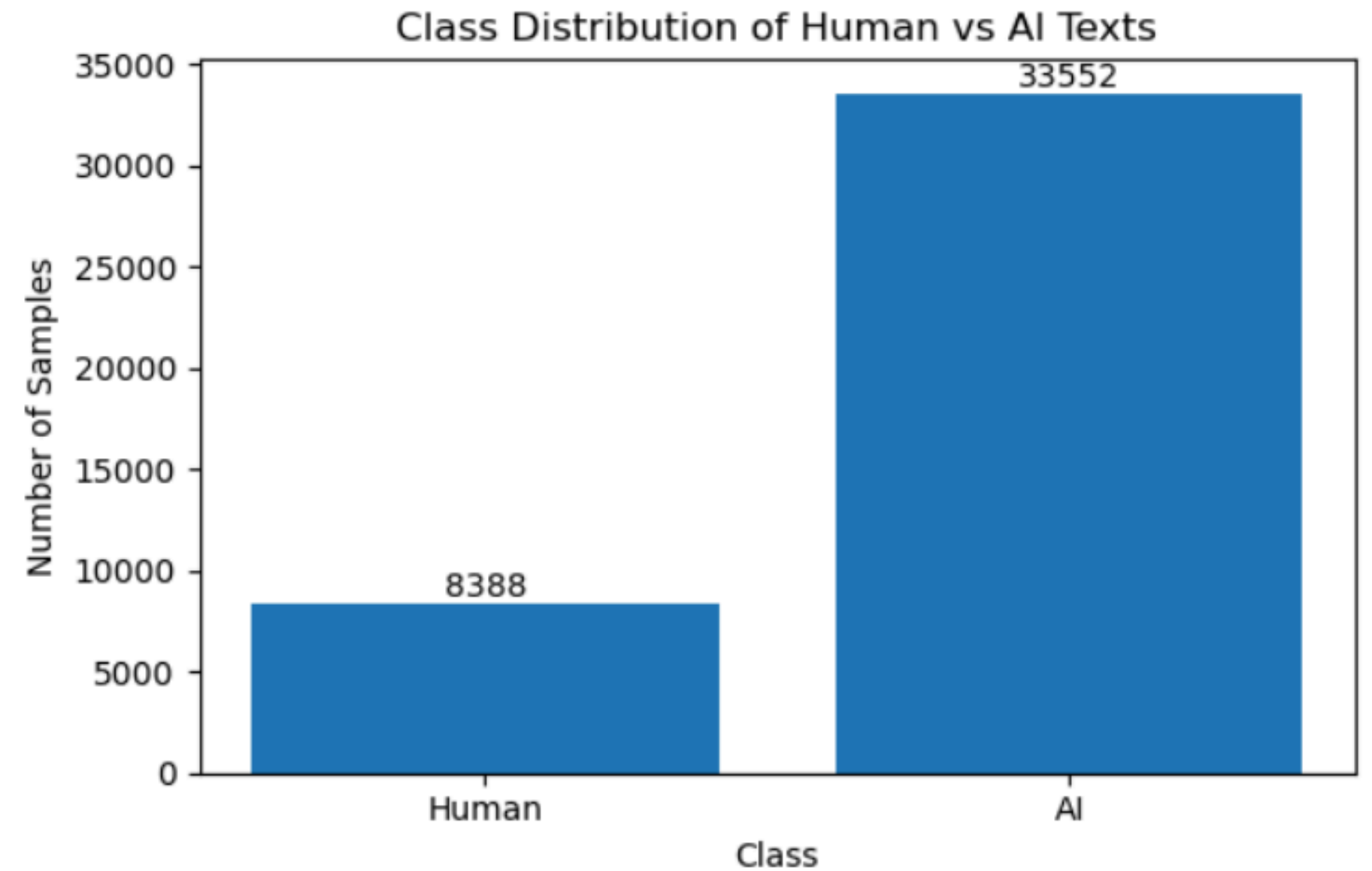
Problem

Motivation

- Rapid growth of AI-generated Arabic content
- Difficulty distinguishing polished AI text from human writing
- Need for reliable detection models

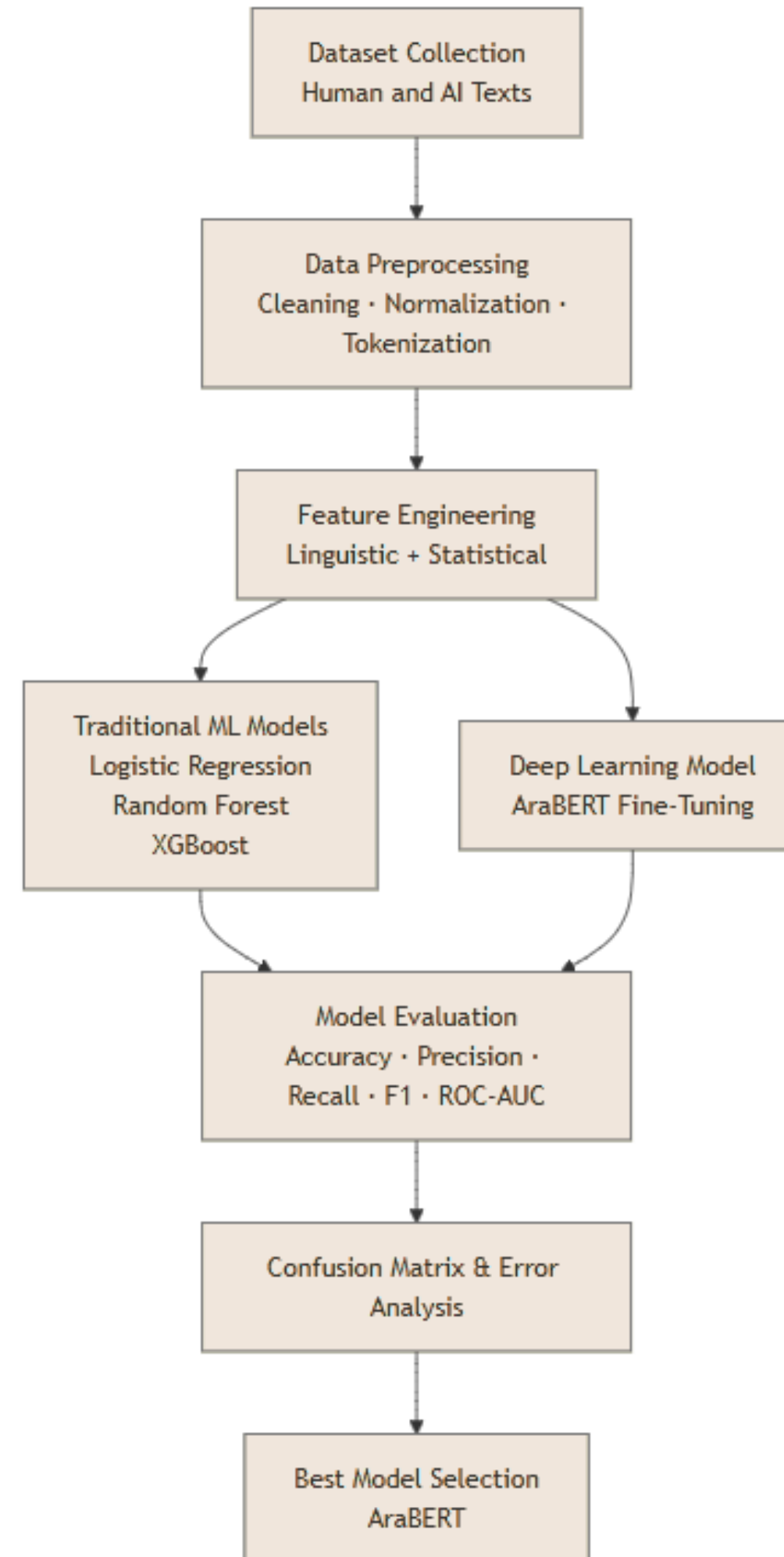
Dataset Overview

- Arabic human & AI-generated texts
- Binary classification task
- Preprocessed and split into train/test sets



Methodology

- Text preprocessing
- Feature engineering (traditional models)
- Model training & evaluation
- Error analysis



Performance Comparison

Models Used

- Logistic Regression
- Random Forest
- XGBoost
- AraBERT (fine-tuned)

Performance Comparison

- Random Forest best traditional model
- XGBoost struggled with minority class
- AraBERT achieved highest overall performance

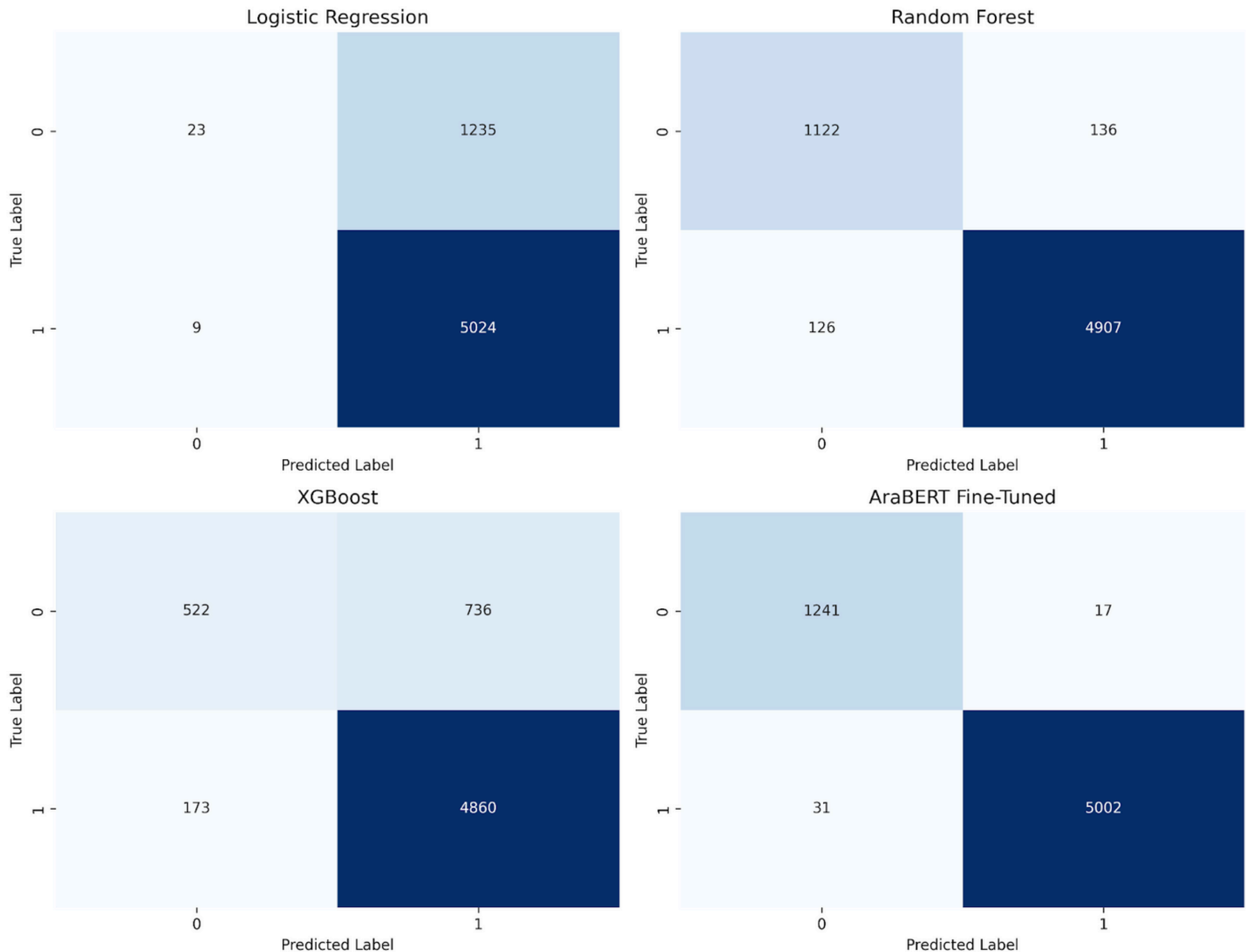
Final Model Results (All Models):

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.80226	0.80268	0.99821	0.88983	0.65666
1	Random Forest	0.95835	0.97303	0.97496	0.97400	0.96691
2	XGBoost	0.85551	0.86848	0.96563	0.91448	0.88710
3	AraBERT Fine-Tuned	0.99285	0.99780	0.99325	0.99552	0.99902

Best Model Based on Highest F1-Score:

	Value
Model	AraBERT Fine-Tuned
Accuracy	0.992847
Precision	0.997804
Recall	0.993245
F1-Score	0.995519
ROC-AUC	0.999018

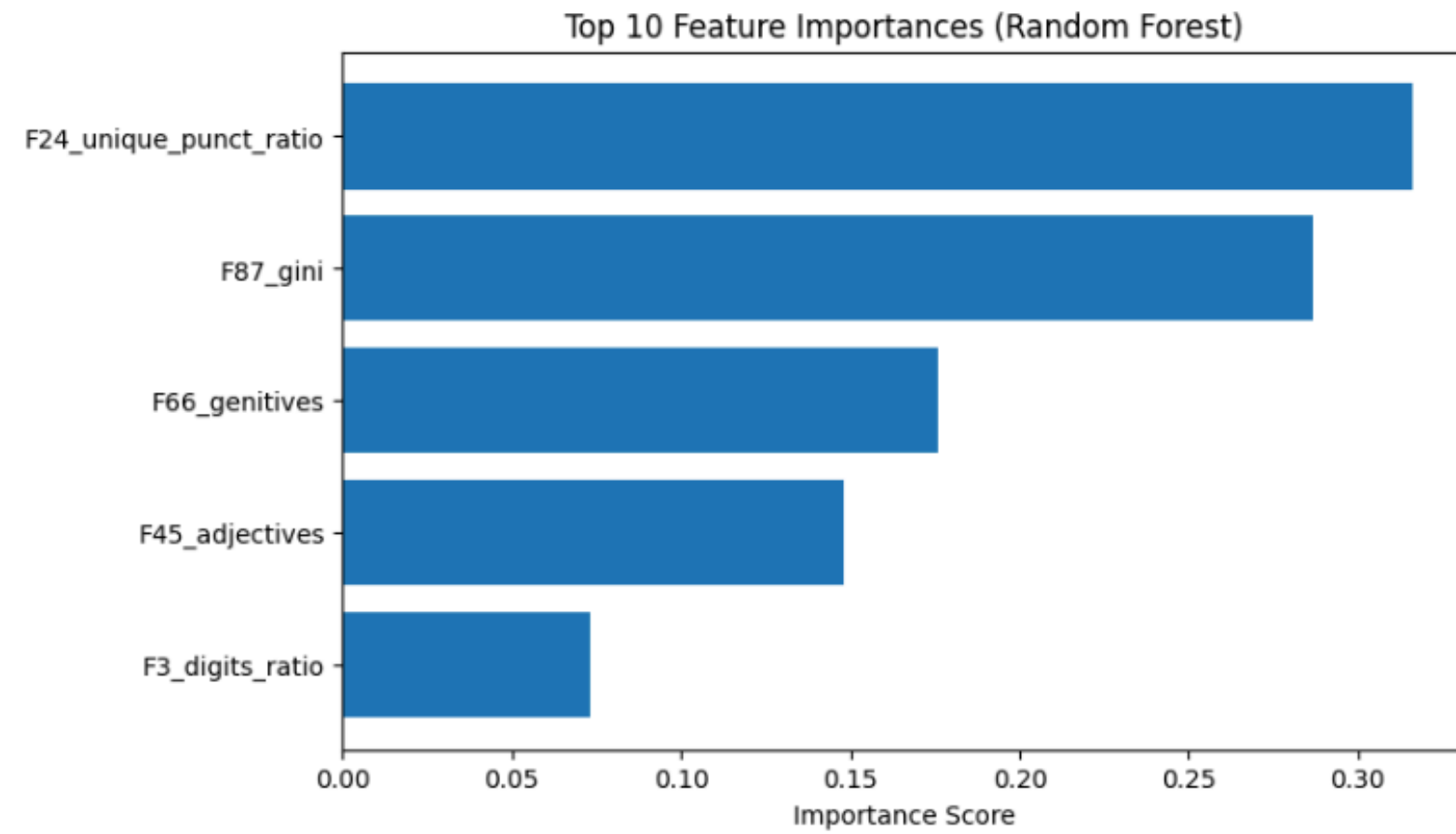
Confusion Matrices Comparison (Test Set)



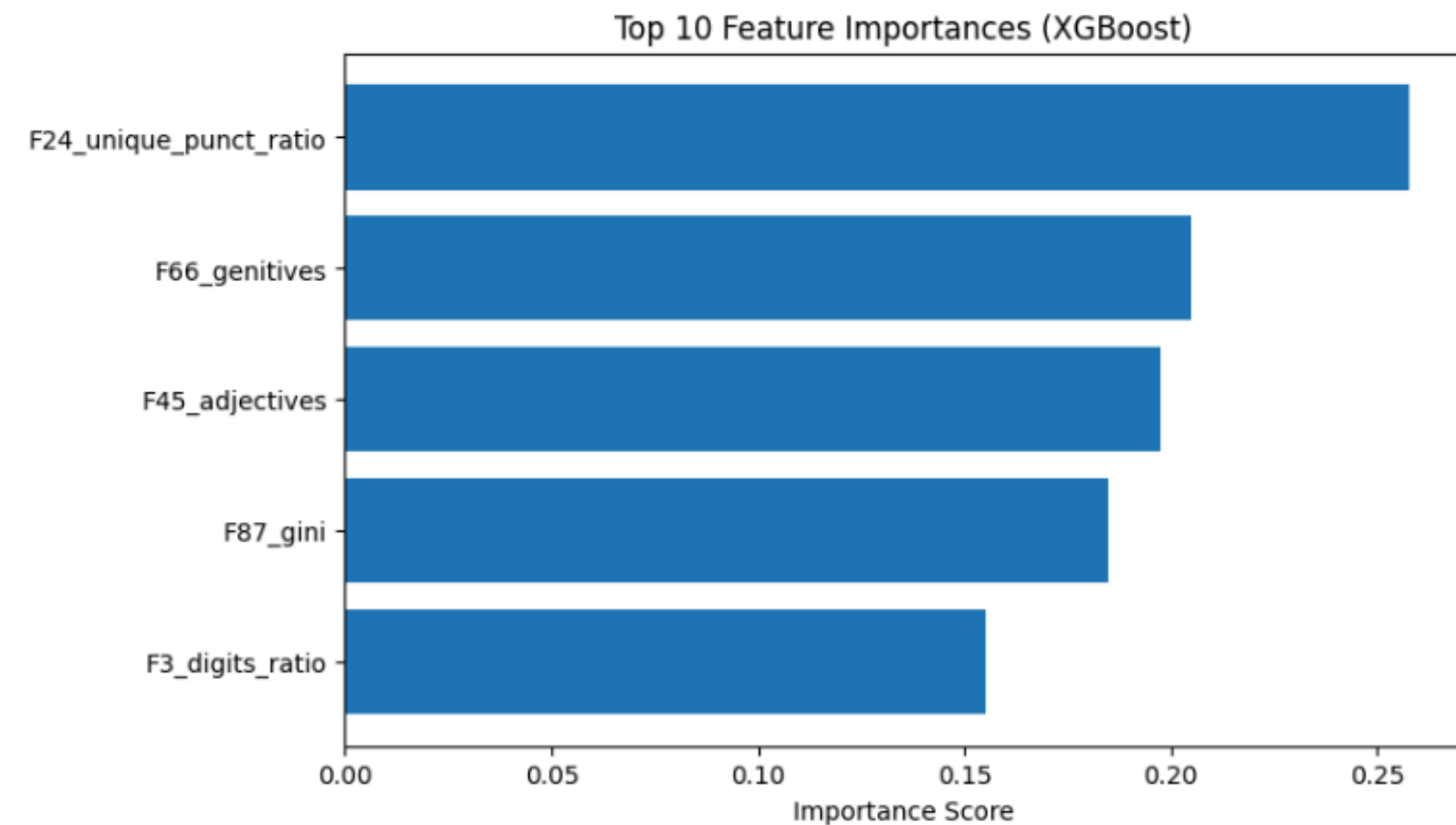
Confusion Matrix Analysis

- Traditional models show more misclassification
- AraBERT reduces both false positives and false negatives

Feature Importance (Traditional Models)



- **F24_unique_punct_ratio** is the most important feature.
- It strongly affects Random Forest and XGBoost decisions.
- AI-generated texts show more consistent and uniform patterns.
- Human-written texts show higher variation.

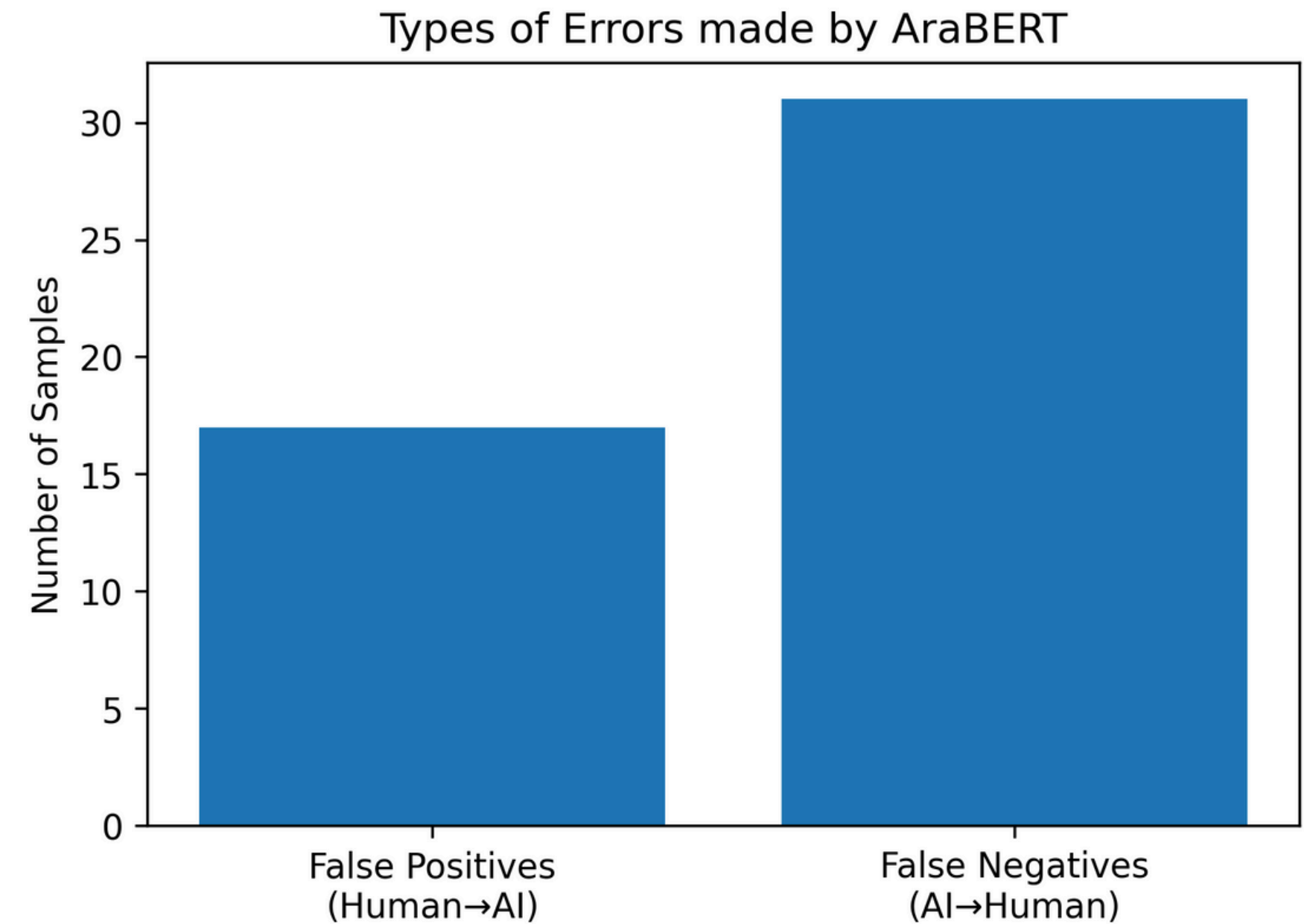


Error Analysis (AraBERT)



Errors occur in:

- Highly polished human text
- Short texts with limited context
- AI- text look like humans writing



Conclusion

- AraBERT is the best-performing model
- Understanding context is very important
- Future work: hybrid models & richer datasets

Thank You!

For your listening...